

SOLAR CELLS

Long-term stability in perovskite solar cells through atomic layer deposition of tin oxide

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Robust contact schemes that boost stability and simplify the production process are needed for perovskite solar cells (PSCs). We codeposited perovskite and hole-selective contact while protecting the perovskite to enable deposition of SnO_x/Ag without the use of a fullerene. The SnO_x , prepared through atomic layer deposition, serves as a durable inorganic electron transport layer. Tailoring the oxygen vacancy defects in the SnO_x layer led to power conversion efficiencies (PCEs) of >25%. Our devices exhibit superior stability over conventional p-i-n PSCs, successfully meeting several benchmark stability tests. They retained >95% PCE after 2000 hours of continuous operation at their maximum power point under simulated AM1.5 illumination at 65°C. Additionally, they boast a certified T_{97} lifetime exceeding 1000 hours.

Despite surpassing the power conversion efficiency (PCE) of many conventional thin-film solar technologies (1–4), perovskite solar cells (PSCs) struggle to achieve long-term stability because of fragile interfaces (5–8). Some contacts degrade under the combination of various environmental stressors, such as humidity, oxygen, temperature changes, and light exposure, which diminishes cell performance and life span (9–13). In inverted p-i-n PSCs, the perovskite–electron transport layer (ETL) interface notably contributes to efficiency losses, forming deep trap states (14, 15). The high cost and poor mechanical properties associated with fullerene ETLs drive the search for alternatives.

However, replacing fullerenes with alternatives, such as metal oxides, has been challenging (16, 17). The thermal sensitivity of hybrid perovskites is problematic, with high-temperature annealing often used for optimal metal oxide processing. Although sputtered oxides damage the perovskite surface, atomic layer deposition (ALD) for SnO_x directly on perovskite creates chemical reactions and interfacial barriers that lead to devices with PCEs below 1% (18–20).

In this work, we redesigned an inverted PSC device architecture with a codeposited perovskite and hole-selective contact deposited as active layers directly on ITO/glass (where ITO is indium tin oxide). The SnO_x prepared through ALD served as a durable inorganic ETL without use of a fullerene. To eliminate detrimental

chemical reactions at the perovskite/ SnO_x interface (21), we introduced aliphatic amine-functionalized perylene-diimide (PDINN) to 3-fluoro-phenethylammonium iodide (m-F-PEAI) surface passivation to provide a one-step passivating and buffering interlayer compatible with ALD deposition.

To enhance device performance, we manipulated the interlayer between the active layers and SnO_x by introducing trace amounts of oxygen vacancies to activate interfacial carriers and facilitate electron extraction, which resulted in a PCE up to 25.1%. The devices also demonstrate excellent stability compared with conventional p-i-n PSCs. Even without encapsulation, the devices perform exceptionally well in various International Summit on Organic and Hybrid Photovoltaics Stability (ISOS) tests, with aging attenuation amplitude less than one-tenth of typical p-i-n devices. Under a prolonged maximum power point tracking (MPPT) stability test at 65°C, the device exhibits a T_{95} aging lifetime exceeding 2000 hours (with third-party certification of a T_{97} stability lifetime surpassing 1000 hours).

Deposition of SnO_x

The device structure of the PSCs consisted of a simple stack of glass/ITO/active layer/passivation layer/ SnO_x/Ag (Fig. 1A). In this structure, the active layers are composed of $\text{Cs}_{0.05}\text{FA}_{0.95}\text{PbI}_{2.94}\text{Br}_{0.06}$ (where FA is formamidinium) doped with [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz), which was prepared by a one-step chemical deposition that has been shown to be effective for constructing inverted PSCs (22). The ETL SnO_x was deposited on the active layer with ALD.

To achieve efficient electron transport, we modulated the tetrakisdimethylamino tin(IV) (TDMASn) to water dosing ratios [TDMASn dose (ms): H_2O dose (ms); 25:20, 50:20, 100:20, 150:20, 200:20, and 250:20] and deposition cycles (from 50 to 500 cycles). As shown by x-

ray photoelectron spectroscopy (XPS) in 1B, figs. S1 to S3, and table S1, SnO_x at 25:20 ratio was closest to the stoichiometric ratio of SnO_2 . As the TDMASn: H_2O ratio increased, the x value of SnO_x gradually decreased.

We further studied the device performance at different TDMASn: H_2O ratios with varying deposition cycles (Fig. 1C and figs. S4 to S12), and the passivation layer was also optimized (figs. S13 and S14). The investigation revealed champion performance with an ALD TDMASn: H_2O dose ratio of 50:20 and a deposition of 300 cycles. Specifically, glass/ITO/active layers/300 cycles SnO_x (Sn:O=50:20)/Ag, achieved a PCE of 21.8% [as shown in fig. S15; open-circuit voltage (V_{OC}) of 1.16 V, short-circuit current density (J_{SC}) of 25.5 mA/cm^2 , and fill factor (FF) of 73.7%].

We noticed a FF deficit for the device using SnO_x as the ETL compared with the C_{60} -based device (fig. S16). A potential cause of the FF loss is the transport resistance, although we found no notable difference between them by comparing the conductivity of C_{60} and SnO_x in fig. S17. Moreover, ultraviolet-visible (UV-vis) spectroscopy and ultraviolet photoelectron spectroscopy (UPS) in figs. S18 and S19 showed that the conduction band minimum (CBM) of SnO_x matched that of the active layers better compared with commercial C_{60} , which suggests that the FF loss was independent of the energy misalignment between the ETL and the perovskite active layer.

In addition to energy level alignment and transport capabilities, the interface between the perovskite and ETL can also determine device performance by affecting carrier extraction efficiency and interfacial defect chemistry. In this case, we further engineered the interface of SnO_x in contact with the perovskite. Before preparing the SnO_x ETL, we deposited an interlayer, named $\text{SnO}_{x(i)}$, with different TDMASn: H_2O ratios and cycle numbers for 10 to 50 cycles. When the TDMASn: H_2O ratio of the $\text{SnO}_{x(i)}$ interlayer is 200:20 and the deposition 30 cycles (Fig. 1D and figs. S20 and S21; specific process curve shown in fig. S22), the device achieved the highest PCE of 25.1%.

The current density–voltage (J - V) characteristics of the champion device (Fig. 1E) with reverse and forward scans yielded PCEs of 25.1% (V_{OC} of 1.19 V, J_{SC} of 25.8 mA/cm^2 , and FF of 81.8%) and 24.1% (V_{OC} of 1.19 V, J_{SC} of 25.7 mA/cm^2 , and FF of 78.9%), respectively. The corresponding stable power output reached up to 24.95% (as shown in fig. S23). Corresponding external quantum efficiency (EQE) spectra in fig. S24 yielded integrated J_{SC} with negligible variation from the values obtained from J - V measurements. Moreover, the process was reproducible, as can be seen from the statistical distribution of all photovoltaic parameters for 20 devices in fig. S25. The performance comparison of

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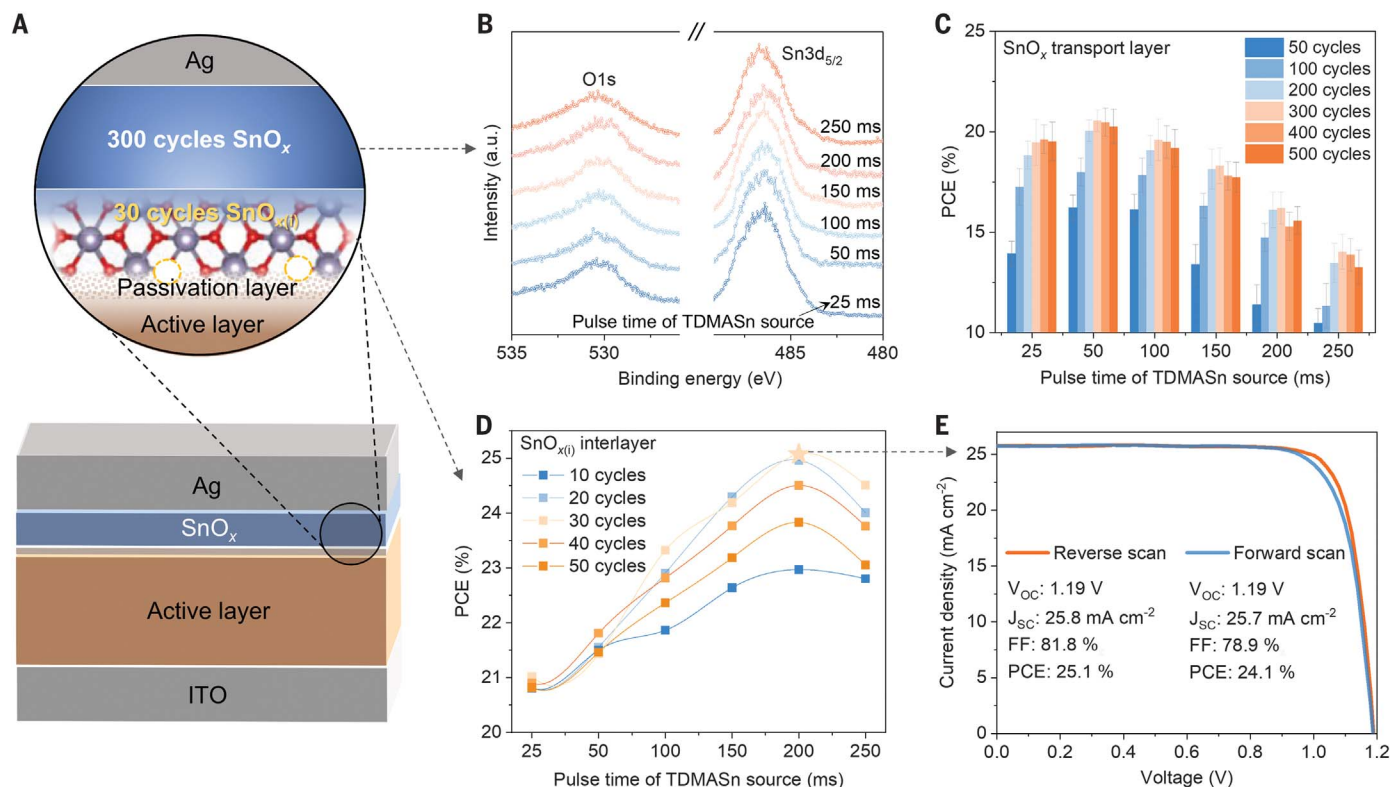


Fig. 1. Device fabrication and performance. (A) Schematic illustration of the device structure. (B) XPS spectra of SnO_x films obtained using different ALD dosing ratios of $\text{TDMASn}:\text{H}_2\text{O}$, with H_2O fixed at 20 ms. a.u., arbitrary units. (C) The impact of SnO_x transport layer on device performance under different ALD dosing ratios of $\text{TDMASn}:\text{H}_2\text{O}$ (H_2O fixed at 20 ms) and different ALD cycles. (D) The

effect of $\text{SnO}_{x(i)}$ interlayer on device performance under different ALD dosing ratios of $\text{TDMASn}:\text{H}_2\text{O}$ (H_2O fixed at 20 ms) and varied ALD cycles. (E) J - V curves of the best-performing SnO_x -based device with the structure of ITO/active layer/passivation layer/30 cycles $\text{SnO}_{x(i)}$ (ALD dosing ratio of $\text{TDMASn}:\text{H}_2\text{O}$ is 200:20) interlayer/300 cycles SnO_x (ALD dosing ratio of $\text{TDMASn}:\text{H}_2\text{O}$ is 50:20)/Ag.

different structural devices involved in this study is shown in fig. S26.

Interlayer characterization

To study the specific effects of the interlayer, we performed electroluminescence quantum efficiency (EQE-EL) and defect density measurements on the devices obtained by depositing the $\text{SnO}_{x(i)}$ interlayer with different $\text{TDMASn}:\text{H}_2\text{O}$ ratios, as well as photoluminescence (PL) characterization on the active layer/ETL films (Fig. 2A). Compared with the device without the interlayer, the EQE-EL value increased with the increase of the $\text{TDMASn}:\text{H}_2\text{O}$ ratio of the $\text{SnO}_{x(i)}$ interlayer, from 3.9% at the $\text{TDMASn}:\text{H}_2\text{O}$ ratio of 25:20 to 11.0% at the $\text{TDMASn}:\text{H}_2\text{O}$ ratio of 200:20 (fig. S27), which we attributed to reduced nonradiative recombination and energy loss (23–25).

Moreover, the trap-filled limit voltage (V_{TFL}) and PL intensity showed opposite trends. Both decreased gradually with the increase of the $\text{TDMASn}:\text{H}_2\text{O}$ ratio of the $\text{SnO}_{x(i)}$ interlayer (specific results are shown in figs. S28 and S29), which suggests that the oxygen-deficient condition in the $\text{SnO}_{x(i)}$ interlayer suppressed defects and promoted carrier extraction (26–28). Additionally, PL mapping on the SnO_x /active layers film and the $\text{SnO}_x/\text{SnO}_{x(i)}$ /active layers

film, respectively (Fig. 2, B and C), revealed a wide PL intensity distribution for the SnO_x /active layers indicative of uneven charge extraction. By contrast, the introduction of the $\text{SnO}_{x(i)}$ interlayer led to a decrease in PL intensity, and the film showed more uniform PL emission, which further verifies that the carrier extraction in $\text{SnO}_x/\text{SnO}_{x(i)}$ /active layers was improved and homogenized (29–32). The surface morphology of the films treated with SnO_x and $\text{SnO}_{x(i)}$, the cross section image of the device, and the thickness of the SnO_x and $\text{SnO}_{x(i)}$ films are shown in figs. S30 to S33.

Theoretical studies

We performed density functional theory (DFT) calculations on various interfaces involving SnO_x and perovskite. The calculations were based on the uncharged heterostructures at the level of Perdew-Burke-Ernzerhof (PBE) + spin-orbit coupling (SOC) as implemented in Vienna Ab initio Simulation Package (VASP) 6.4 (33–35). The results are presented in Fig. 2, D to K, and the detailed data are shown in figs. S34 to S37, where we compared the charge distribution and partial density of states (pDOS) at the interfaces, both with and without the $\text{SnO}_{x(i)}$ (V_{O}) slab and the PDINN buffering layer (where V_{O} refers to the oxygen vacancy).

The selection of both pristine SnO_2 and $\text{SnO}_{x(i)}$ (V_{O}) with oxygen vacancies is to clarify the effect of the formation of oxygen vacancies (i.e., $\text{TDMASn}:\text{H}_2\text{O}$ ratios from 25:20 to 200:20) on the charge carriers. For the PDINN molecule, which may give very large supercell and strong lattice mismatch, we keep only one amino side chain instead of the two side chains in the original designation—on the other side, we adopt a methyl group instead.

The external-adjacent Sn^{4+} and I^- usually have a high degree of chemical reactivity, especially when they are directly combined with hetero components at interface. The formation of V_{O} structure (Fig. 2E) can be considered a result of partial reduction of Sn^{4+} cations (36), which exhibits higher spatial charge density distribution at the conducting band (CB) edge compared with the pristine SnO_2 (Fig. 2D). With the strong interaction between the perovskite and SnO_2 layers, the fiercely distorted Pb-I frameworks tend to directly bind with the SnO_2 layer (as shown in fig. S34). As such, the interfacial perovskite structures bonded with SnO_2 have the same CB edge level as SnO_2 (Fig. 2, D and H, and figs. S35A and S36A), removing the band offset between SnO_2 and perovskite layers while weakening the carrier transfer trend between the interface.

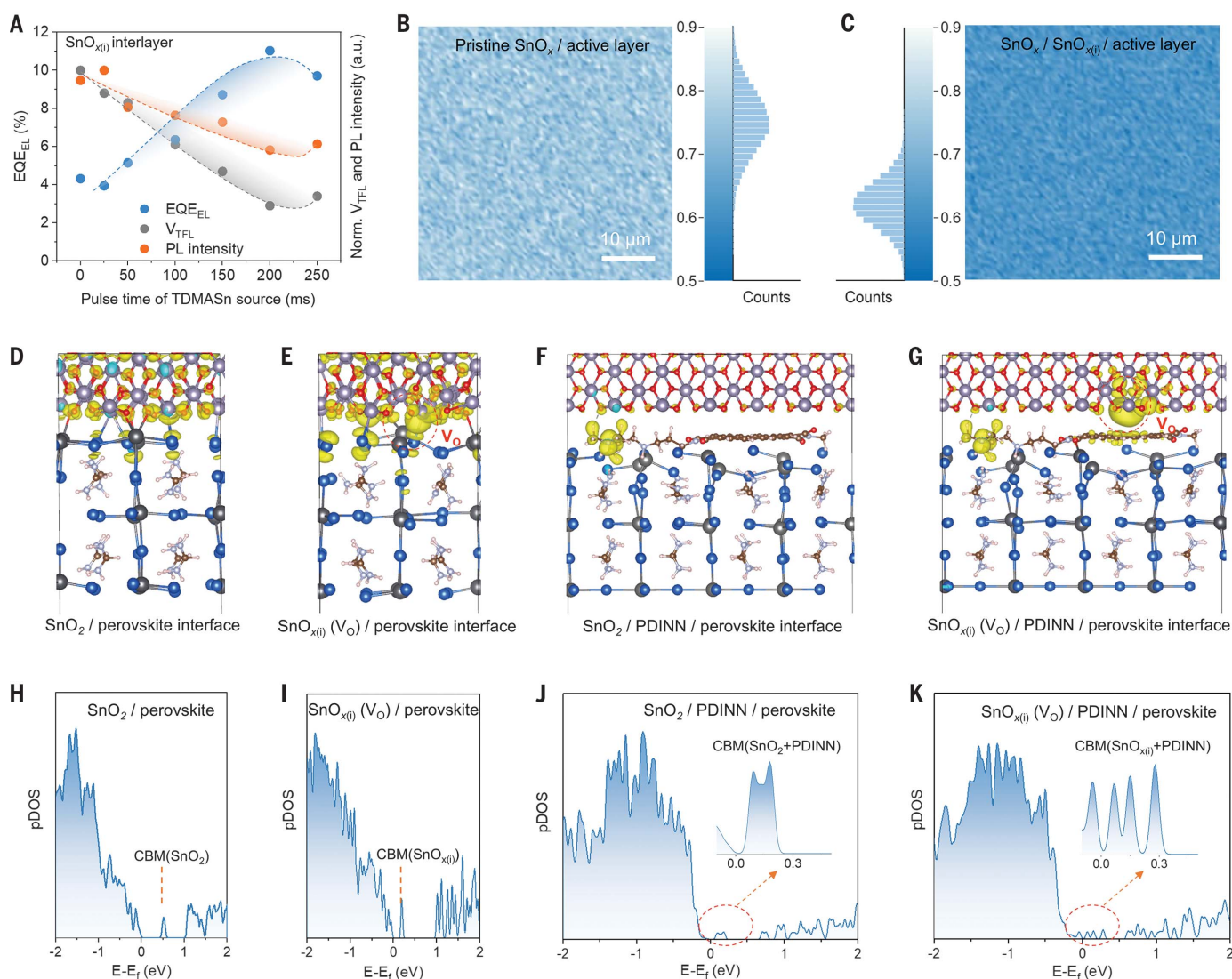


Fig. 2. Interlayer modulation. (A) EQE-EL, V_{TFL} , and PL intensity of the $\text{SnO}_{x(i)}$ interlayer obtained with different pulse time of TDMA source (the pulse time of H_2O sources is fixed at 20 ms). (B and C) PL mapping (incident from the SnO_x side) of the films, including the pristine SnO_x /active layer (B) and the $\text{SnO}_x/\text{SnO}_{x(i)}$ /active layer (C). (D to G) Illustration of the simulated charge distribution (corresponding to CBM) at different interfaces, including the pristine

SnO_2 /perovskite interface (D), the $\text{SnO}_{x(i)}(\text{V}_\text{O})$ /perovskite interface (E), the SnO_2 /PDINN/perovskite interface (F), and the $\text{SnO}_{x(i)}(\text{V}_\text{O})$ /PDINN/perovskite interface (G). (H to K) Computed pDOS plots of the pristine SnO_2 /perovskite interface (H), the $\text{SnO}_{x(i)}(\text{V}_\text{O})$ /perovskite interface (I), the SnO_2 /PDINN/perovskite interface (J), and the $\text{SnO}_{x(i)}(\text{V}_\text{O})$ /PDINN/perovskite interface (K).

Additionally, with the V_O defects, the strong surface interaction further induces the band-gap downshift between the CB edge of $\text{SnO}_{x(i)}$ and the valence band (VB) edge of the perovskite (Fig. 2, E and I, and figs. S35B and S37A).

With the introduction of a PDINN molecular buffer layer (as shown in Fig. 2F), it is remarkable that there is no effective spatial charge distribution at the interface within the planar polycyclic aromatic backbone of PDINN in the vacancy-free structure while only partial effective spatial charge distribution arises at the amino side chains. Similarly, the pDOS in Fig. 2J and figs. S35C and S36 shows a substantial decrease in the DOS peak at the edge of the CBM, indicating a reduced possibility of carrier dis-

tribution in this region. This result suggests that polycyclic aromatic backbones of PDINN do not promote charge transport but rather act as a buffering agent for the ALD process owing to its hydrophobic properties, which can prevent H_2O from directly contacting and reacting with the halide perovskite as demonstrated by XPS analyses in fig. S38.

For the V_O -existence $\text{SnO}_{x(i)}/\text{PDINN}/\text{perovskite}$ interface (Fig. 2G), the presence of oxygen vacancies leads to an appearance of spatial charge density contribution of $\text{SnO}_{x(i)}$ adjacent with the polycyclic aromatic part of PDINN. Similarly, the CBM peaks in pDOS (Fig. 2K and figs. S35D and S37) exhibit stronger dispersion than those in the vacancy-free bench-

mark. Without the direct contact between the $\text{SnO}_{x(i)}$ and perovskite layers due to the PDINN layer, the strongly dispersed CBM peak ranges result in the reduction of the negative “cliff-like” band offset compared with the benchmark (Fig. 2, F and J), which further reduces the energy loss during carrier extraction and transfer (37–39). These first-principles computational results collectively suggest that the appearance of the V_O defects increased the carrier transfer, thereby activating the $\text{SnO}_{x(i)}/\text{PDINN}/\text{perovskite}$ interface by inducing carrier distribution into the polycyclic aromatic backbone parts of the PDINN layer. This conclusion is also applicable to the mF-PEAI and PDINN dual-additive interface because PDINN

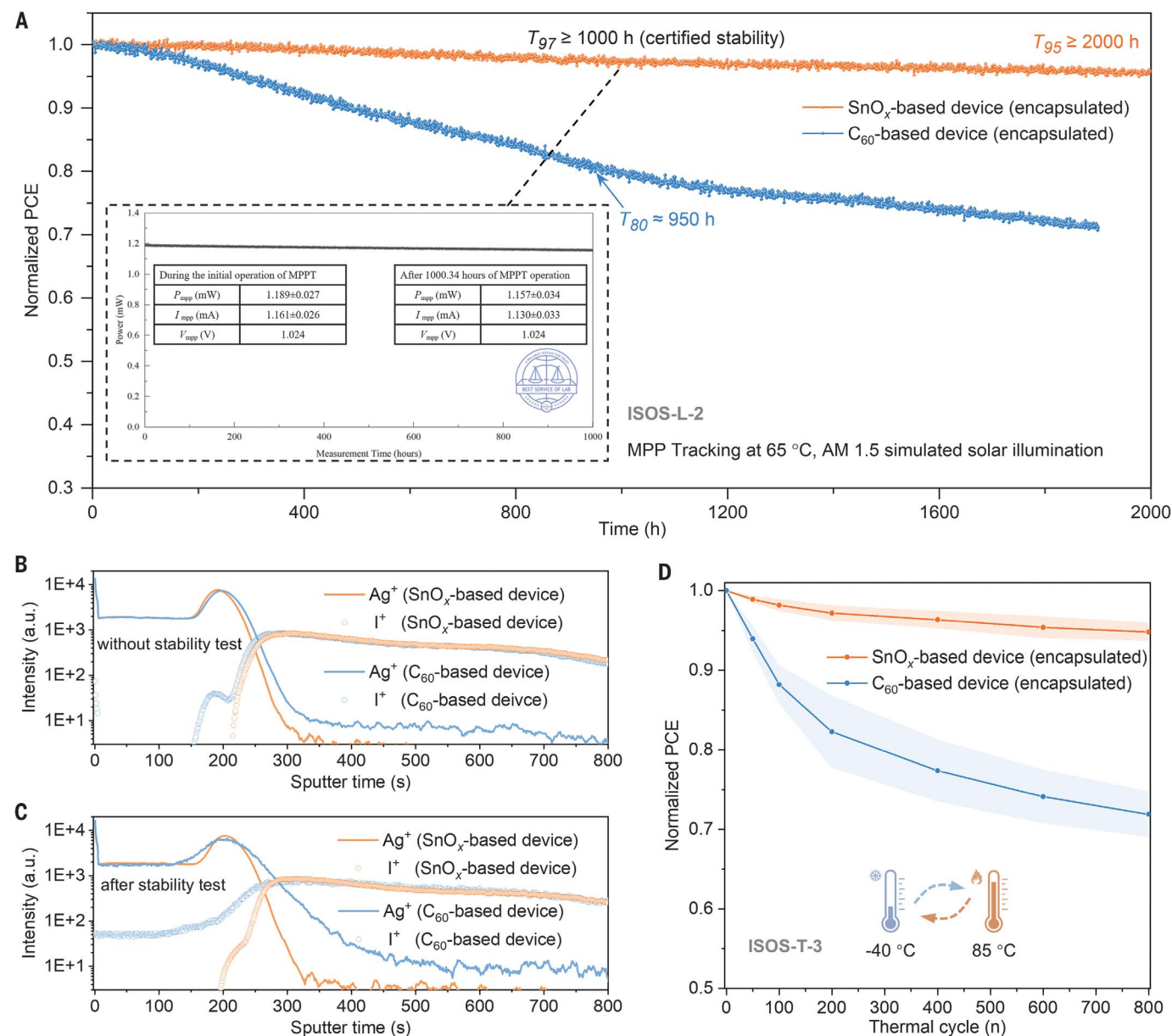


Fig. 3. Operational stability. (A) Operational stability of devices at temperature of 65°C. (B) TOF-SIMS characterization of original device without stability test. (C) TOF-SIMS characterization of device after the operational stability test in (A). (D) Devices' efficiency evolution under repeated thermal cycling (−40°C to 85°C) in the dark in air (ISOS-T-3). We conducted thermal cycling tests using eight individual devices and obtained the average performance change along with the SD as the error measure.

dominates the charge density contribution owing to its stronger conjugation effect compared with the isolated single aromatic ring of mF-PEAI, as proven in fig. S39.

Operational stability

The encapsulated devices were placed in air under continuous 1-sun illumination (following the ISOS-L-2 protocol), and the performance of the devices was monitored at 65°C under MPPT. The SnO_x-based devices show excellent stability, maintaining >95% of the initial PCE after 2000 hours of continuous operation ($T_{95} \geq 2000$ hours; Fig. 3A). Moreover, to obtain certified stability results, the SnO_x-based devices

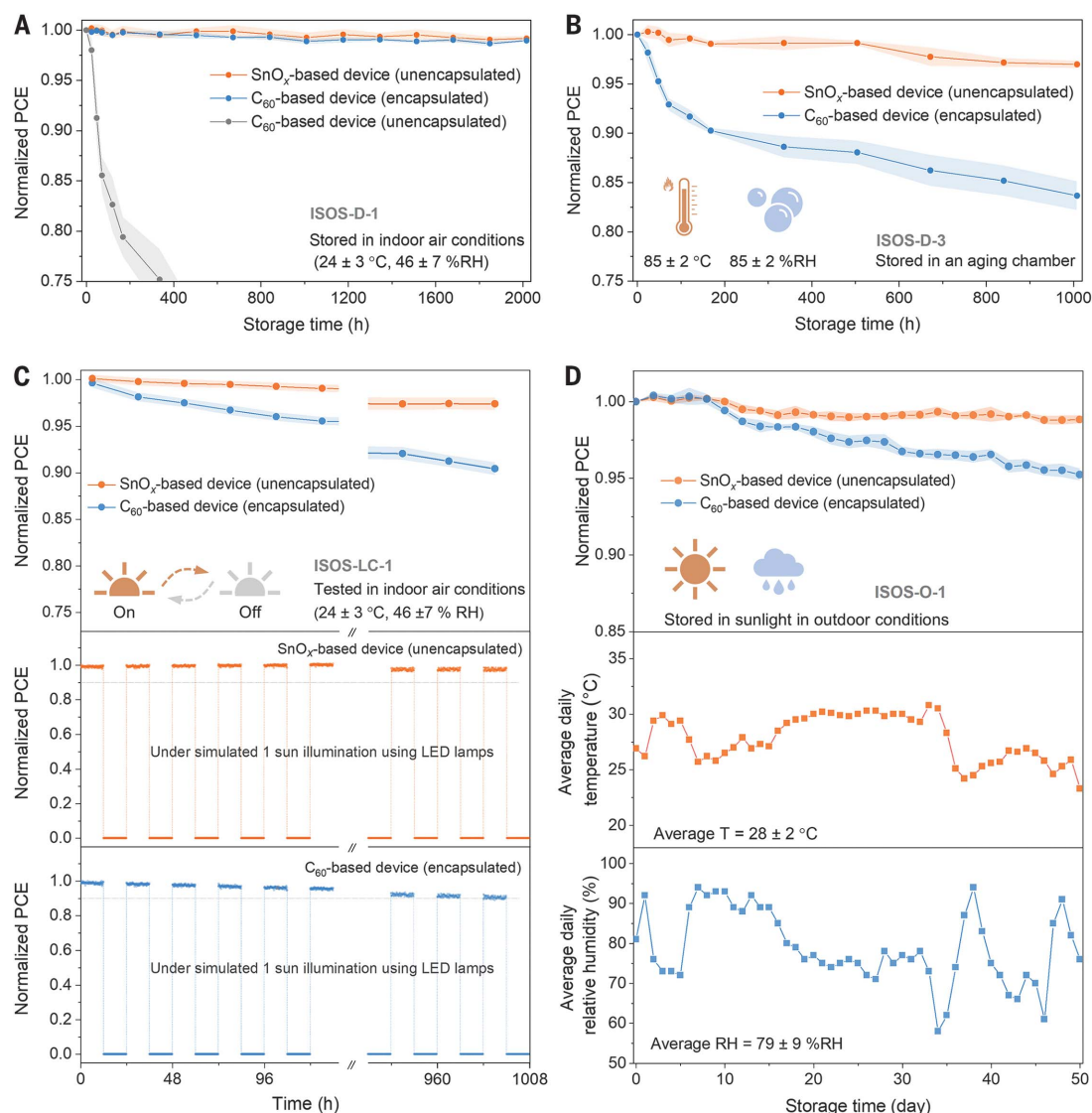
were sent to a third-party institution for stability testing and demonstrated an exceptional stability of >97% of the initial PCE after 1000 hours of continuous operation (shown in the dashed box in Fig. 3A and fig. S40). Table S2 details a favorable comparison with other recently reported stability results. By contrast, the control device maintains a relatively stable performance in the first 100 hours, but the PCE dropped to 80% of the initial value after 950 hours of continuous operation. Similar trends were observed even during accelerated testing at 85°C and under 1-sun illumination (fig. S41).

To gain insight into the difference in device operational stability, we performed time-of-flight

secondary ion mass spectrometry (TOF-SIMS) tests on the devices before and after the stability tests. In Fig. 3, B and C, for the untested control device, a small amount of I[−] ions penetrated into the C₆₀/BCP layer and contacted the Ag electrode. After the stability test, the diffusion of I[−] ions intensified, manifested as a large number of I[−] ions penetrating into the C₆₀/BCP layer and entering the Ag electrode (figs. S42 and S43). For the SnO_x-based device, under the protection of SnO_x, I[−] ions did not show any obvious diffusion in the device before and after aging (figs. S44 and S45). This result indicates that the photothermal stability of the SnO_x-based device comes from the

Fig. 4. Environmental stability.

(A) Devices' efficiency evolution under storage in dark under indoor air conditions (ISOS-D-1). **(B)** Devices' efficiency evolution under damp heat testing at 85°C and 85% RH in the dark in air (ISOS-D-3). **(C)** Devices' efficiency evolution under light on-off cycle test (12 hours–12 hours) using light-emitting diode (LED) lamps simulated 1-sun illumination (ISOS-LC-1). **(D)** Devices' efficiency evolution under storage in sunlight under outdoor condition (ISOS-O-1). (The devices are in an open-circuit state. Environmental data are quoted from the Hong Kong Observatory, <https://www.hko.gov.hk/tc/wxinfo/pastwx/mws2023/mws202309.htm>.) We conducted stability tests for each condition [(A) to (D)] using 10 to 14 individual devices and obtained the average performance change along with the SD as the error measure.



protection of the SnO_x layer to the perovskite layer, which makes it difficult to undergo ion diffusion and potential phase separation under the dual influence of high temperature and light.

The thermal cycling test, which is a common test used by the International Electrotechnical Commission (IEC) for outdoor conditions and commercialization, was also performed. For the test (−40°C to 85°C, dark; ISOS-T-3), the SnO_x-based devices showed an average degradation of ~4.7% after 800 thermal cycles (as shown in Fig. 3D), which exceeded the 200 thermal cycles usually performed by this test protocol. The control devices showed an average efficiency reduction of 28.1% after experiencing the same conditions.

Environmental stabilities were studied on the SnO_x-based PSCs and conventional inverted PSCs (referred to as control devices) according to multiple sets of standard environmental stability tests, including ISOS-D-1, ISOS-D-3,

ISOS-LC-1, and ISOS-O-1. For the ISOS-D-1 stability test shown in Fig. 4A and fig. S46A, after storing for 2000 hours at 23° ± 4°C and 46 ± 7% relative humidity (RH), both encapsulated and unencapsulated SnO_x-based devices showed almost no performance degradation and maintained >99.1% of their initial PCE for 2000 hours. The unencapsulated control device exhibited obvious performance degradation (the PCE dropped to 76.4% of the initial value after 300 hours), whereas the encapsulated control device maintained 98.9% of the initial PCE after storing for 2000 hours. The result indicates that the SnO_x layer can effectively diminish water and oxygen damage around the perovskite, effectively achieving self-encapsulation of the devices.

We further performed high-temperature and high-humidity storage (ISOS-D-3) at 85°C and 85% RH. As shown in Fig. 4B, after storing for 1000 hours, the unencapsulated SnO_x-based device showed a decay of <3.0%. Moreover, encapsulating it did not appreciably improve

its stability (the attenuation was reduced from 3.0% to 2.1% in fig. S46B). In comparison, the encapsulated control device showed a decay of >16.4%, indicating that external encapsulation alone was insufficient to resist the invasion of extreme environments. In addition, we performed light on-off stability experiments (ISOS-LC-1) in indoor environments. As shown in Fig. 4C, after 42 cycles of 12 hours–12 hours light on-off test at 23° ± 4°C and 46 ± 7% RH, the unencapsulated SnO_x-based device showed a PCE decay of 2.5%, whereas the encapsulated control device attenuates by 9.7%. We further studied the stability of the devices under outdoor aging conditions (ISOS-O-1).

Outdoor aging tests

Because outdoor temperature, relative humidity, and solar radiation intensity all affect the stability of the devices, we tracked the changes of these parameters during the outdoor aging test (Fig. 4D and fig. S47). After placing in the

outdoor environment for 50 days, the unencapsulated SnO_x -based device shows a decay of only 1.1%, indicating that the self-encapsulated SnO_x -based device is almost unaffected by the outdoor environment. In comparison, the encapsulated control device shows a relatively large attenuation of 4.8%. The stability results validate the self-encapsulation mechanism of SnO_x devices. It inhibits external moisture and oxygen penetration, extending device lifetime. It also suppresses ion diffusion from the perovskite layer to the ETL or electrode, maintaining device performance (fig. S48A). By contrast, control devices with an additional encapsulation layer experience adhesive aging, potentially leading to encapsulation failure (fig. S48B). The photos of the original device and the encapsulated device are shown in fig. S49. The encapsulation layer above the electrode fails to prevent ion diffusion that may occur during long-term operation.

Discussion

We demonstrated the effectiveness of a simplified fullerene-free PSC device structure featuring SnO_x by modulating oxygen vacancy defect concentration in the interlayer. We reveal that the fine-tuning of oxygen defects reduces the negative cliff-like band offset between perovskites and ETLs, thereby decreasing energy losses during carrier extraction and transfer. The modulated device exhibits excellent efficiency up to 25.1% and superior stability. The attenuation amplitude of unencapsulated devices in various standard stability tests, such as ISOS-D-1, ISOS-D-3, ISOS-LC-1, and ISOS-O-1, is less than one-tenth of that of traditional p-i-n devices. Under long-term MPPT stability tests at 65°C, the

T_{95} aging life of the device exceeds 2000 hours (third-party certified T_{97} stable life exceeds 1000 hours). This simplified, efficient, and stable device architecture provides insight into the development of low-cost and reproducible PSCs.

REFERENCES AND NOTES

1. H. Chen *et al.*, *Science* **384**, 189–193 (2024).
2. Z. Li *et al.*, *Science* **376**, 416–420 (2022).
3. H. Gao *et al.*, *Science* **383**, 855–859 (2024).
4. National Renewable Energy Laboratory, "Best Research-Cell Efficiency Chart"; <https://www.nrel.gov/pv/cell-efficiency.html>.
5. H. Zhu *et al.*, *Nat. Rev. Mater.* **8**, 569–586 (2023).
6. L. Duan *et al.*, *Nat. Rev. Mater.* **8**, 261–281 (2023).
7. B. Li *et al.*, *Chem* **10**, 35–47 (2024).
8. S. Zhang *et al.*, *Adv. Energy Mater.* **10**, 2001610 (2020).
9. P. Chen *et al.*, *Nature* **625**, 516–522 (2024).
10. S. M. Park *et al.*, *Science* **381**, 209–215 (2023).
11. Z. Li *et al.*, *Science* **382**, 284–289 (2023).
12. Q. Jiang *et al.*, *Nature* **623**, 313–318 (2023).
13. R. Azmi *et al.*, *Science* **376**, 73–77 (2022).
14. J. Warby *et al.*, *Adv. Energy Mater.* **12**, 2103567 (2022).
15. A. A. Said *et al.*, *Nat. Commun.* **15**, 708 (2024).
16. J. P. Bastos *et al.*, *ACS Appl. Mater. Interfaces* **8**, 9798–9805 (2016).
17. Z. Li *et al.*, *Joule* **2**, 1559–1572 (2018).
18. A. E. A. Bracesco *et al.*, *ACS Appl. Mater. Interfaces* **15**, 38018–38028 (2023).
19. N. Mallik *et al.*, *Nano Energy* **126**, 109582 (2024).
20. A. Hultqvist *et al.*, *ACS Appl. Energy Mater.* **4**, 510–522 (2021).
21. F. A. Palmstrom *et al.*, *Adv. Energy Mater.* **8**, 1800591 (2018).
22. X. Zheng *et al.*, *Nat. Energy* **8**, 462–472 (2023).
23. G. Yang *et al.*, *Nat. Photonics* **15**, 681–689 (2021).
24. J. Zhu *et al.*, *Nat. Energy* **8**, 714–724 (2023).
25. W. Peng *et al.*, *Science* **379**, 683–690 (2023).
26. J. Wang *et al.*, *Nat. Commun.* **11**, 177 (2020).
27. Q. Jiang *et al.*, *Nat. Energy* **2**, 16177 (2016).
28. D. Yang *et al.*, *Nat. Commun.* **9**, 3239 (2018).
29. Y. Bai *et al.*, *Science* **378**, 747–754 (2022).
30. C. Liu *et al.*, *Nat. Commun.* **12**, 6394 (2021).
31. A. Krishna *et al.*, *Energy Environ. Sci.* **14**, 5552–5562 (2021).
32. T. Zhang *et al.*, *Joule* **2**, 2706–2721 (2018).
33. G. Kresse, *J. Non-Cryst. Solids* **192–193**, 222–229 (1995).
34. P. E. Blöchl, *Phys. Rev. B* **50**, 17953–17979 (1994).
35. J. P. Perdew, K. Burke, M. Ernzerhof, *Phys. Rev. Lett.* **77**, 3865–3868 (1996).
36. Q. Liu, M.-G. Ju, W. Liang, *Phys. Chem. Chem. Phys.* **22**, 20553–20561 (2020).

37. H. Chen *et al.*, *Nature* **613**, 676–681 (2023).
38. Z. Yu *et al.*, *Nat. Energy* **5**, 657–665 (2020).
39. M. Stollerfoht *et al.*, *Energy Environ. Sci.* **12**, 2778–2788 (2019).

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SUPPLEMENTARY MATERIALS

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Materials and Methods

Figs. S1 to S49

Tables S1 and S2

References (40–57)

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